Sensors & Actuators: B. Chemical 272 (2018) 282–288

Contents lists available at ScienceDirect

Sensors and Actuators B: Chemical

journal homepage: www.elsevier.com/locate/snb

Machine learning enabled acoustic detection of sub-nanomolar concentration of trypsin and plasmin in solution

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ARTICLE INFO

Keywords:
Casein
Plasmin
Trypsin
Protease
QCM
Biosensor
Machine learning

ABSTRACT

We demonstrate a machine learning enabled low-cost acoustic detection of protease which may find application in assuring quality and safety of dairy products, drug screening, molecular profiling, and disease diagnostics. A hydrophilic SiO_2-coated quartz crystal microbalance (QCM) acts as a substrate to assemble α-, β-, and κ-casein layers (protease reporters) and as a transducer for measuring changes in frequency as casein is removed by protease. We demonstrate that α-, β-, and κ-caseins can form stable assembly on SiO_2 from phosphate-buffered solution (PBS) solution. Exposure to protease results in cleaving of casein which changes the frequency of the 1st–11th odd harmonics of QCM. Monitoring β-casein cleavage allows ~0.2 nM detection of trypsin and ~0.5 nM detection of plasmin and enables differentiation between trypsin and plasmin after < 2 min of protease exposure. The casein-coated QCM allows sub-nanomolar detection and classification of protease.

1. Introduction

Proteases are involved in broad range of biological processes including digestion of long protein chains (pepsin), blood-clotting (thrombin, plasmin), metabolic control (elastase) as well as high tonnage green bioprocesses in waste water treatment, bio-fuel and biopolymers production and dairy industry. Cascade biochemical reactions triggered by protease lead to induction or repression of the expression of genes, subsequently leading to negative changes on the cellular level. Development of low cost methods for assessing protease activity are critically important for unfolding the complexity of protease activity in biological systems development of in situ diagnostics for industrial processes, and diagnostics for preventive health care. An example of the most prevalent endogenous protease present in bovine milk, plasmin, may retain up to 40% of its original activity in milk after ultra-high temperature treatment which is meant to neutralize protease activity [1,2]. Plasmin-induced cleaving of bonds in the casein molecule causes collapse of the casein micelle followed by aggregation of proteins which negatively affects milk texture and flavor [2]. Monitoring plasmin activity is essential for controlling the quality of dairy products [3]. Conventional methods of plasmin detection (on the level of 1 nM plasmin detection in undiluted plasma) are time-consuming, expensive and utilize antibodies which constrain plasmin activity by binding to its active site and forming inactive plasmin-inhibitor complexes [4,5]. Plasmin was detected at 0.6 nM concentration by following a decrease in the value of ferrocene redox current after cleavage of the peptide-substrate by plasmin [6] Castillo et al. further improved this approach to detect 0.56 nM plasmin in milk [7]. Sensitive fluorometric tests were developed for detecting plasmin activity in the tears of patients with eye diseases like microbial keratitis, cornea ulcer and post-traumatic herpetic infection [8]. Bell et al. developed a fluorometric assay for plasmin based on measurement of intensity of fluorescence of β-naphthol released from α-N-methyl-α-N-tosyl-I-lysine β-naphthol ester after cleaving by plasmin, which required a stoichiometric plasmin-streptokinase complex for detection of < 0.2% concentration [9]. A surface plasmon resonance (SPR) technique was used by Ledoux et al. to probe plasmin interaction with substituted dextran heparin-like polymer RG1192 with < 10 nM plasmin detection limit [10]. However, few reported plasmin detection techniques are practical for widespread use because they require complex protein synthesis [11], preparation of cell cultures [12,13], or radioactive materials such as technetium–99 m for labelling [14,15].

Design of novel plasmin biosensors capable of detecting low-mass cleavage of proteins, distinguishing between proteolysis caused by different mechanisms, and scalability for industrial use is of significant interest [16]. The quartz crystal microbalance (QCM) [17] is an

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https://doi.org/10.1016/j.snb.2018.05.100
Received 19 February 2018; Received in revised form 16 May 2018; Accepted 17 May 2018
Available online 19 May 2018
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acoustic technique which presents an attractive platform for development of a gravimetric plasmin biosensor because of its sub-nanogram resolution [18], simultaneous measurement of mass change and viscoelastic properties [19], label-free analyte detection [20] and scalability [21]. Krause et al. demonstrated QCM coated with natural fibrinogen and artificial short peptide for detection of human neutrophil elastase (HNE) and cathepsin detection at 0.55 Uml^{-1} level [22,23]. The detection of 0.65 nM plasmin using β-casein layer immobilized on a mercaptoundecanoic acid (MUA) on an 8 MHz QCM was recently reported by Poturnayova et al. [24] While the detection limit was low, synthesis and immobilization of the peptide substrate may be prohibitively labour and time intensive for low cost application [6,24].

Here we demonstrate that caseins, the most common milk proteins, self-assembled on the surface of SiO_2 and can be used for label-free detection of plasmin activity by acoustic biosensors based on multi-harmonic response of a quartz crystal microbalance. Caseins in milk exist in the form of pseudo-micelles composed of different casein types that play a unique role in protein binding and transport of amino acids, calcium and phosphorous.

We assembled α-, β-, and κ-caseins on a hydrophilic SiO_2 surface without the use of cross-linkers, which often cause undesirable detection artifacts during protease sensing. Mass change (Δm) of a thin rigid casein layer deposited on a QCM crystal is related to frequency shift (Δf) of the crystal by $\Delta f \propto n\ \Delta m$, where n is the harmonic number of the crystal [17]. Since frequency shift in response to a given mass change is proportional to the harmonic number, signal-to-noise ratio and detection sensitivity increase with crystal harmonic.

"To quantify the different in the dynamics of casein adsorption on the surface of the QCM we used Support vector machines (SVM). SVM belong to machine learning statistical techniques designed to enable computers to progressively improve classification/categorization of data. SVM is based on Vapnik-Chernovenkis theory of computational learning [25]. We used SVM Classification Learner application in Matlab (MathWorks) to differentiate between α-, β-, and κ-casein during self-assembly on the surface of QCM crystal. Casein deposition time and frequency shifts of the 1 st–11th odd harmonics of the QCM crystal were used as predictor variables and casein type (α, β, or κ) was used as the response variable. We found that a support vector machine algorithm with quadratic kernel (quadratic function) allows to distinguish between α-, β-, and κ-casein, while linear kernel was not suitable for classification due to nonlinear decision boundaries.

We further classified changes in multi-frequency response of κ-casein coated QCM crystal to protease during proteolysis using Support Vector Regression (SVR), implemented using the scikit-learn library in python environment [24]. Using multi-harmonic QCM frequency response, we investigated assembly and stability of self-assembled α-, β-, and κ-casein layers on SiO_2 and measured the dynamics of QCM-response during casein cleaving by protease. Performance of the casein-based biosensor for differentiation between plasmin and trypsin as well as classification of caseins based on their kinetics of adsorption on SiO_2 was assessed. Since pH change can induce desorption of casein [26,27], we investigated stability of the assembled casein biosensing layers while pH of the buffer solution was changed.

2. Experiment

Pre-cleaned Si-coated AT-cut quartz crystals with resonant frequency 5 MHz (International Crystal Manufacturing Co., Inc., USA) were plasma-treated (Diener Zepto plasma system, Germany) in air for 15 min. After plasma treatment, crystals were immediately inserted in the flow cell of a KSV QCM-Z500 system (Finland). Resonant frequency and parasitic resistance of the crystals were measured under 50 μL/min flow of deionized water. The water was then replaced by 10 mM phosphate-buffered (10 mM Na_2HPO_4, 2 mM KH_2PO_4, 2.7 mM KCl and 137 mM NaCl) pH 7.4 (PBS) solution at 50 μL/min flow rate. PBS was filtrated by a 0.22 μm membrane filter (Merck-Millipore, Germany) prior to use. To measure kinetics of casein assembly on SiO_2, fundamental and overtone frequencies of the QCM were measured while PBS solution with 0.05 mg/mL concentration of α- (containing α_{s1} and α_{s2} with ratio 3.8:1 by wt.), β-, and κ-casein (Sigma-Aldrich, USA) was flowed over the SiO_2-coated crystal surface until resonant frequency of the crystal stabilized, signaling completion of assembly. The low casein concentration was used to allow growth of a single monolayer without binding of aggregates or micelles. PBS solution without casein was then flown over the crystals to remove loosely-bound casein and test stability of the deposited layers. For proteolysis measurements, solutions with 1 nM and 10 nM concentrations of trypsin, thrombin and plasmin in PBS were flowed over QCM crystals with immobilized β-, and κ-casein layers at 50 μL/min flow rate. Trypsin and thrombin were used as obtained from Sigma-Aldrich. Plasmin was freshly prepared before each experiment according to the procedure reported by Castillo et al. [8]. During deposition and cleaving of casein layers, frequency shifts of the 1st, 3rd, 5th, 7th, 9th, and 11th harmonics were recorded every 10 s using a KSV QCMZ-500

For pH stability measurements, a small amount of HCl or NaOH was added to the PBS solution. The pH of each solution was measured using an Accumet Excel XL60 pH meter (Fisher Scientific, USA) calibrated using 3 commercial solutions of pH 4, 7, and 9. Solutions were exchanged in the QCM flow cell immediately after preparation due to the pH instability of PBS outside its working pH range. Solutions were then flowed across an un-cleaved β-casein layer deposited on SiO_2 using the method previously described. Each pH-altered solution was applied to the β-casein layer until frequency of the QCM reached stable values, generally observed over a period of 3–4 h.

Molecular dynamic simulations of the assembly of 25 α_{s1} casein molecules into micellar structures were conducted using NAMD [28] at the following conditions: the structures were equilibrated in water at 0.2 molar density (total 5962 water molecules) with explicit counter ions accounting for charge-neutrality of the simulation box. All the simulations were performed in NPT ensemble thereby allowing change in the simulation cell size. Periodic boundary conditions in 3D were used for a box of length 11.74 nm x 12.11 nm x 11.96 nm. The final structures were produced from a 2 ns simulation after the equilibration of 1 ns. While there is no direct correlation between the dielectric constant and pH, a lower dielectric constant generally represents stronger electrostatic interactions akin to higher pH and vice versa. Molecular dynamic simulation of twenty-five molecules of α_{s1} casein showed that the protein final morphologies depend on solution acidity, Fig. S1(a, b, c). At lowest pH (Fig. 1S (a)), the casein molecules are separated (a condition close to the isoelectric point) while at pH higher than isoelectric point (D = 1.0) the 25 casein molecules form one large micelle. (Fig. S1 b and c).

Classification of casein was carried out using the Classification Learner application in Matlab (MathWorks). A support vector machine algorithm with quadratic kernel (quadratic SVM) was used to distinguish between α-, β-, and κ-casein based on the QCM response during casein self-assembly on SiO_2. A linear kernel was not used for classification due to nonlinear decision boundaries between α-, β-, and κ-casein. Casein deposition time and frequency shifts of the 1st–11th odd harmonics of the QCM crystal were used as predictor variables and casein type (α, β, or κ) was used as the response variable. For classification of protease during proteolysis of κ-casein, a support vector regression (SVR) algorithm was implemented using the scikit-learn library in python [29]. A decision boundary between frequency shifts acquired during trypsin exposure and those acquired during plasmin exposure was constructed by training the SVR using 50% of the data and validating the model using 50% of the data by 5-fold cross-validation in order to avoid overtraining to the high-harmonic noise. Support vector algorithms were used for regression/classification because of their simplicity, ease in visualization, and established use in biosensing applications [30,31]. Support vector hyperparameter were not changed from their default values in Matlab or scikit-learn.

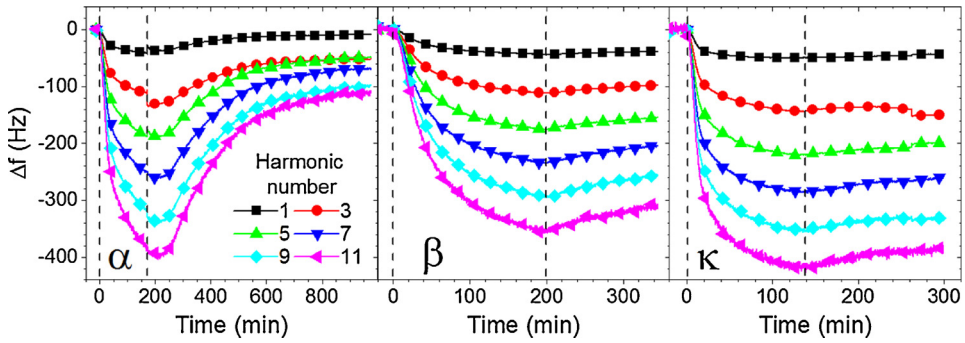

Fig. 1. Kinetics of casein adsorption and removal from SiO_2, as measured by frequency shift (Δf) of 1st–11th odd harmonics of the QCM. Exposure and removal of α-casein (left panel), β-casein (middle panel), and κ-casein (right panel) are marked by the first and second vertical dashed lines respectively.

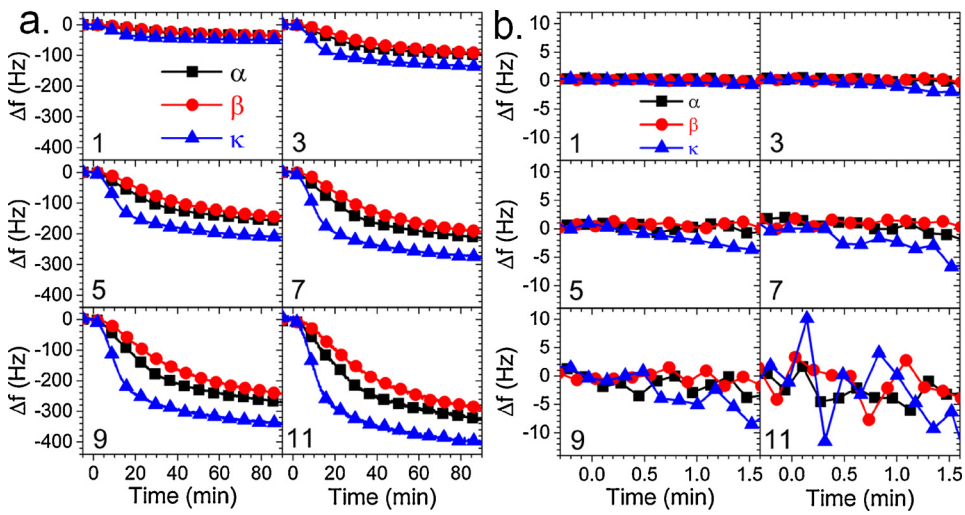

Fig. 2. Resonant frequency shifts of 1st-11th odd harmonics during deposition of α-, β-, and κ-casein for 90 min (a) and for 1.5 min (b). Harmonic number is shown in the lower left corner of each panel. Casein exposure occurs at time = 0 min.

3. Results and discussion

Immobilization and removal of casein from SiO_2. Frequency shifts in 1st, 3rd, 5th, 7th, 9th, and 11th harmonics of SiO_2-coated QCM crystals are shown in Fig. 1 during exposure to solutions containing α-casein (left panel), β-casein (middle panel), and κ-casein (right panel). In each case, exposure to casein is marked by the first vertical dashed line (at time = 0) and removal of casein by PBS is marked by the second vertical dashed line. The crystals exhibited a sharp drop in resonant frequency upon exposure to α-, β-, and κ-casein, indicating adsorption of casein on the SiO_2 surface. Stabilization of the fundamental resonant frequency (1st harmonic) occurs 100–200 min after exposure to casein. The low casein concentration in PBS (0.05 mg/mL), which is below the critical micellar concentration for casein, allows proper self-assembly of casein molecules on the SiO_2 surface and results in slow formation of immobilized layers. Changes in the QCM frequency response during casein deposition are enhanced at higher overtones, which results in about 350–450 Hz frequency shift in resonant frequency of the 11th harmonic. Resonant frequency increases after flushing by PBS (second vertical dashed line), but does not return to the original resonant frequency measured before crystals were exposed to casein. This suggests that only loosely-bound casein layers are removed during the PBS washing. Casein adsorption on SiO_2 results in formation of a tightly-bound monolayer and reversibly-bound second layer which can be removed by flushing with a buffer [32]. The increase in resonant frequency observed after PBS flushing likely corresponds to removal of the second adsorbed layer.

As indicated by the shift in fundamental resonant frequency (Δf = −40 Hz) after α-casein adsorption and washing by PBS (Δf = −9 Hz), nearly 80% of the α-casein was removed by PBS flushing. The primary role of α-casein as a molecular chaperone in milk makes it more important for supporting the formation of other casein-based structures than for creating stable structures on its own, which likely prevents immobilization of stable α-casein layers on SiO_2. During PBS washing of β-casein, only a small increase in fundamental frequency was observed, which suggests that the majority of β-casein adsorbed on SiO_2 formed stable immobilized layers. From shift of the fundamental resonant frequency and the Sauerbrey equation [17], we estimate that the immobilized β-casein layer has mass density 350 ± 100 ng/cm^2. This is in good agreement with measurements by Tiberg et al., who predicted β-casein monolayer formation with density 246 ng/cm^2 and observed formation of 433 ng/cm^2 films using ellipsometry and neutron reflection [33]. It is likely that β-casein adsorption results in an asymmetric bilayer rather than a pure monolayer. After formation of a dense monolayer, a self-assembled second layer is formed which may contain unstable regions that are removed during washing by PBS. Adsorption of κ-casein on SiO_2 resulted in a larger frequency shift than that of α- and β-caseins, likely because of denser bilayer packing of κ-casein due of its low molecular weight and stronger hydrophilicity than α- and β-caseins [34]. Structure of the κ-casein layer is likely similar to the casein micelle structure, with the fibrillar moiety facing the external surface of the film, which enhances stability of the layer during washing with PBS.

Differentiation of α-, β-, and κ-casein during adsorption on SiO_2. We investigated the possibility of rapid identification of α-, β- and κ-casein based on kinetics of their adsorption on SiO_2. Fig. 2a shows frequency

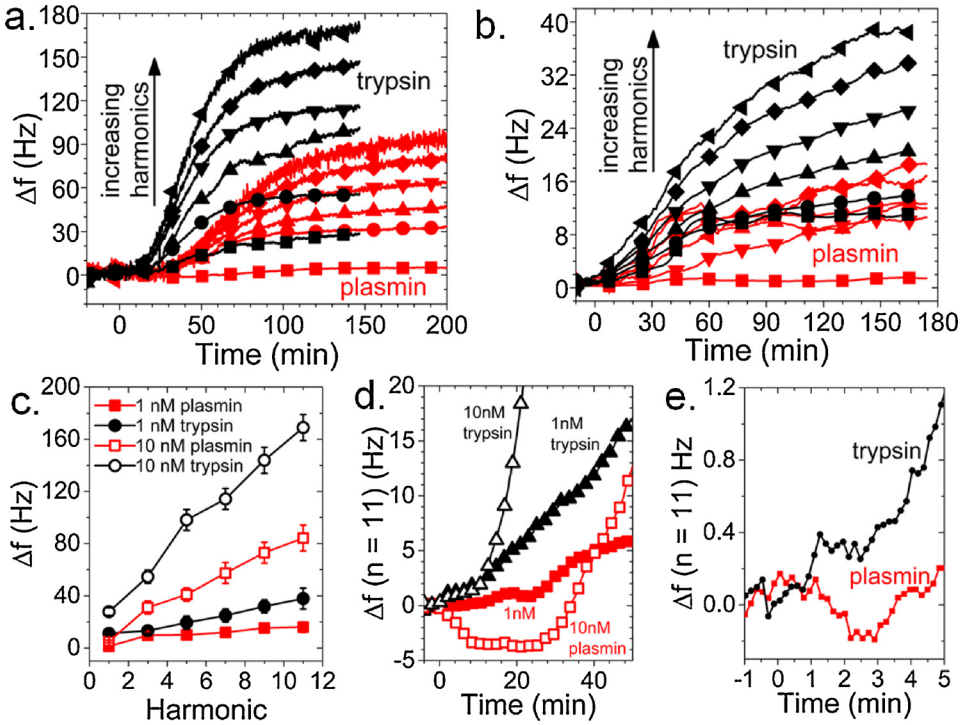

Fig. 3. a. Multi-harmonic frequency shifts of QCM crystals coated with immobilized β-casein in response to (a) 10 nM and (b) 1 nM trypsin and plasmin protease in PBS. Protease exposure occurs at time = 0 min. (c) Frequency shift vs. crystal harmonic after 140 min of 10 nM and 1 nM protease exposure. Error bars correspond to amplitude of noise in (a) and (b). (d) Frequency shift of the 11th crystal harmonic upon exposure to 10 nM and 1 nM protease. (e) Frequency shift of 11th crystal harmonic during exposure to 1 nM trypsin and plasmin showing the time scale (< 2 min) required for clear differentiation between trypsin and plasmin.

shifts of 1st-11th odd harmonics of SiO_2-coated QCM crystals during the first 90 min of casein assembly. At high (7th-11th) harmonics, it is possible to clearly distinguish between α-, β-, and κ-caseins due to differences in kinetics of their self-assembly on SiO_2. While α- and β-casein assembly results in frequency shift that can be described by a single exponential decay with time constant τ, κ-casein assembly more closely resembles bimodal behavior (τ_1 and τ_2). However, similarities in the observed adsorption rates make distinction between caseins difficult. Moreover, the time required to reach steady-state QCM response may be prohibitively long for some sensor applications. Significantly shorter detection times are desirable for biosensing of casein in industrial applications. An alternative to direct kinetic fits, which may also be affected by the concentration-dependence of the observed kinetics, is application of statistical approaches, which can significantly reduce the time required for casein identification. Considering that for industrial applications the sensing system should be able to identify caseins in less than 2 min, we replotted frequency shifts of 1st-11th odd harmonics during the first 1.5 min of casein deposition (Fig. 2b). It is clear that direct comparison does not allow simple differentiation between α-, β-, and κ-casein. For statistical treatment, we compiled a casein adsorption matrix containing frequency shifts at 1st, 3rd, 5th, 7th, 9th, and 11th harmonics and the elapsed deposition time. A quadratic SVM was used to construct a hyperplane which acted as a decision boundary to differentiate between caseins in the vector space. A 5-fold cross-validation method was used to prevent overtraining to high noise levels inherent in the high-harmonic frequency shifts. Due to the statistical nature of the SVM results, the SVM was run 5 times and the average classification accuracy was recorded. Classification accuracy for distinguishing between α-, β-, and κ-casein reached 100% using the first 3.5 min of casein deposition time (Fig. S2). After casein deposition for 1 min, identification accuracy was found to be about 92% (highlighted region in Fig. S2), suggesting the possibility of rapid (~1 min) differentiation between α-, β-, and κ-casein during adsorption on SiO_2 using low concentration (0.05 mg/mL) and low flow rate (50 μL/min) conditions.

Proteolysis of immobilized β-casein. We demonstrated stability of immobilized β- and κ-casein films on SiO_2 during PBS washing, which makes them attractive for potential application in biosensors for detection of protease activity. Multi-harmonic frequency shift of QCM crystals coated with immobilized β-casein layers during exposure to trypsin and plasmin are shown in Fig. 3a and b for 10 nM and 1 nM protease concentrations respectively. Protease exposure occurs at time = 0 min. Frequency increases during exposure to 10 nM and 1 nM protease solutions indicate that significant proteolysis occurs at both concentrations, corresponding to cleavage of casein. Frequency shift during trypsin exposure is roughly a factor of 2 higher than that during plasmin exposure for both 10 nM and 1 nM concentrations. Trypsin has at least one more cleavage site on β-casein than plasmin, and trypsin has a higher affinity than plasmin to the cleavage sites that both proteases have in common[35], which results in the high trypsin response in β-casein. During exposure to 10 nM trypsin, β-casein cleavage results in Δf = 27 Hz in the fundamental (n = 1) resonant frequency, corresponding to a loss of roughly 70% of the total mass of the immobilized β-casein film.

Frequency shift of the β-casein film in response to protease exposure is dramatically enhanced when measured at higher harmonics. Frequency shift after 140 min of 10 nM and 1 nM trypsin and plasmin exposure at different crystal harmonics is shown in Fig. 3c. Frequency response of the β-casein film to 10 nM plasmin exposure is enhanced by a factor of 12 (from ~6 Hz to ~84 Hz) when measured from the 1st to 11th harmonic. Frequency shift upon exposure to both trypsin and plasmin increases roughly linearly with crystal harmonic, which demonstrates the importance of using higher harmonic sensing modes for increasing the sensitivity of acoustic-based biosensors. Since the highest detection sensitivity is achieved using the 11th crystal harmonic, we examined proteolysis kinetics at the 11th harmonic in Fig. 3d. At both 10 nM and 1 nM concentrations, proteolysis occurs more rapidly upon interaction with trypsin than plasmin because of the higher number of cleavage sites available to trypsin. As expected, 10 nM trypsin results in more rapid proteolysis than 1 nM trypsin. Exposure to 1 nM plasmin results in slow proteolysis which occurs over the course of ~3 h (Fig. 3b). The initial drop in frequency upon exposure to 10 nM plasmin is likely due to adsorption of plasmin on the bare gold electrode of the QCM before significant proteolysis takes place. This was observed in other studies in which poor coverage of gold by a peptide substrate, confirmed using atomic force microscopy, resulted in plasmin

adsorption on gold [24]. Kinetics of the 10 nM plasmin response suggest that significant proteolysis occurs after ~30 min of plasmin exposure. Differences in frequency response of the 11th harmonic upon exposure to 10 nM protease ($\Delta f_{n=11} < 0$ for plasmin, $\Delta f_{n=11} > 0$ for trypsin) allows rapid (~1 min) differentiation between trypsin and plasmin without the use of statistical tools. At 1 nM protease concentration, clear differentiation between trypsin and plasmin at is possible after ~2 min using response of the 11th harmonic (Fig. 3e). The results demonstrate that self-assembled casein may be used for rapid (< 2 min) detection and identification of protease at nM concentrations. We estimated detection limits of trypsin and plasmin using signal to noise ratio (SNR) and magnitude of the 11th overtone QCM responses shown in Figs. 3a and 3b. After 140 min of protease exposure, resonant frequency of the 11th overtone underwent shifted by 170 ± 10 Hz and 39 ± 3 Hz during exposure to 10 nM and 1 nM trypsin respectively. Uncertainties were estimated from noise in the frequency response and resulted in SNR of ~13.5. The SNR allows for estimated trypsin detection limit of ~0.2 nM concentration. Resonant frequency of the 11th overtone shifted by 86 ± 10 Hz and 19 ± 2 Hz after 140 min of plasmin exposure, resulting in a SNR of ~9. This allows estimated plasmin detection limit of at ~0.5 nM concentration.

To test the possibility of non-specific binding of protease to the β-casein layer, we also measured the β-casein frequency response during exposure to thrombin, which does not cleave β-casein. Application of 10 nM thrombin resulted in a small drop in frequency due to non-specific adsorption, but no significant frequency shift indicating cleavage of β-casein was observed. The lack of response to thrombin demonstrates that immobilized casein exhibits specific response to trypin and plasmin which enables selective detection of these proteases.

pH stability of immobilized β-casein. We have demonstrated that adsorbed β-casein was stable under PBS flow and served as a biosensing layer for distinguishing between trypsin and plasmin activity. Since β-casein interaction with hydrophilic SiO_2 is expected to be affected by concentration of H^+ ions and subsequent changes in pKa of the SiO_2 surface, the effect of pH environment on stability of the casein layer was investigated. pH-dependent changes in the fundamental resonant frequency of the β-casein layer relative to resonant frequency in the neutral 7.4 pH environment are shown in Figure S3. Alkaline pH caused significant desorption of β-casein, resulting in a ~10 Hz frequency shift corresponding to loss of ~25% of film mass at pH 8 and ~15 Hz frequency shift corresponding to loss of ~35% of film mass at pH 9. These results are comparable to β-casein removal reported at similar pH observed using ellipsometry[27]. The removal of β-casein upon exposure to alkaline pH is likely caused by pH-induced increase in net surface charge (pKa) which increases electrostatic repulsion between adsorbed casein molecules and between casein and the SiO_2 surface, which leads to instabilities in assembled casein layers. Resonant frequency shift during exposure to PBS at pH < 6.5 also caused observable frequency increase, but after return to the original pH, frequency returned to its original value. This is in agreement with previous studies showing that small changes in temperature and pH result in reversible changes in cohesion of casein layers [26]. As pH decreases, accumulation of charge on casein increases rigidity/viscosity of the casein layer, reducing drag at the surface of the crystal, and resulting in an increase in resonant frequency. When pH returns to its original value, viscosity of the casein layer decreases, resulting in a decrease in the crystal resonant frequency. We further elucidated the pH dependence structural changes of casein molecules by investigating the α_{s1} casein self-assembly in bulk using molecular dynamics simulation. In a 25 α_{s1} casein system with a hydration level of 0.2 (see Fig. S1 in supplementary information). At the lowest dielectric constant, each casein molecules form segregated clusters of their own, while at higher dielectric constant, all the casein molecules collapse into a large globular structure.

Proteolysis of immobilized κ-casein layer. In natural bovine milk, κ-casein is often concentrated at the surface of the casein micelle where it acts as an interface between the hydrophobic interior of the micelle and the external environment, which exposes it to interaction with protease [36]. To investigate performance of κ-casein as a substrate for protease detection, we measured QCM response during κ-casein interaction with 10 nM trypsin and plasmin (Fig. 4a). Similar frequency shift is observed in response to both trypsin and plasmin, which is slightly (~5 Hz) lower that the response of β-casein to 10 nM trypsin (Fig. 3a). Since the κ-casein layer forms the other shell of casein micelles and is generally considered resistant to proteolytic cleaving [37], it is noteworthy that application of 10 nM trypsin and 10 nM plasmin caused cleavage and removal of ~40% of the mass of the original κ-casein layer. Cleavage may have occurred at the κ-casein site between Phe105 and Met06, as suggested by Huppertz [38]. The similar frequency shifts in response to trypsin and plasmin suggest that cleavage by both proteases occurs at the same site, which may limit the ability of κ-casein-based biosensors to distinguish between trypsin and plasmin activity [35]. Frequency shift of κ-casein after 220 min of protease exposure is shown vs. harmonic number in Fig. 4b. As was observed for β-casein (Fig. 3c), Δf increases roughly linearly with crystal harmonic. The similarity in frequency response to trypsin and plasmin at low harmonics demonstrates that differentiation between proteases using a κ-casein-based acoustic biosensor requires the use of high order (n = 7–11) harmonics that exhibit large frequency response during a given mass change.

We estimated detection limits of the κ-casein-based sensor using SNR of the frequency response to 10 nM plasmin and trypsin. After 240 min of protease exposure, resonant frequency of the 11th overtone shifted by 204 ± 5 Hz and 189 ± 6 Hz during plasmin and trypsin exposure respectively. The frequency response during proteolysis of the immobilized κ-casein enables detection limits of ~0.2 nM plasmin and 0.3 nM trypsin. This is roughly a factor of 3 lower than the 0.65 nM plasmin detection limit of β-casein layer on QCM reported by Poturnayova et al., which utilized the fundamental harmonic of an 8 MHz crystal [24].

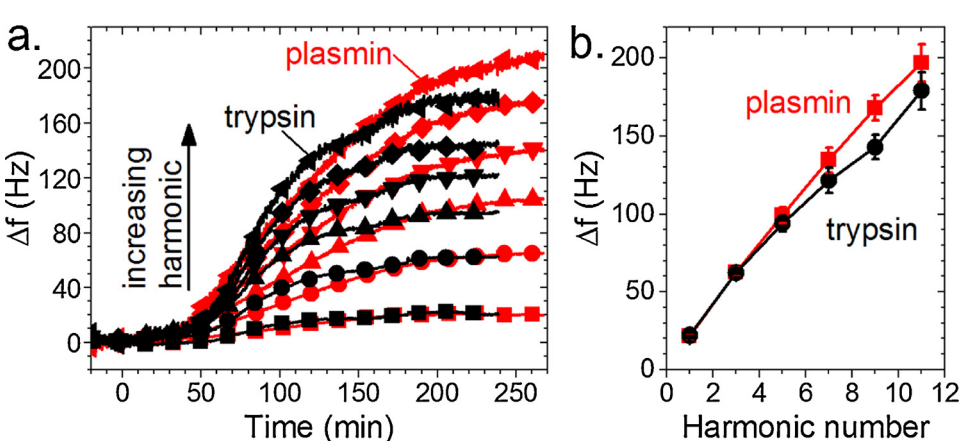

Fig. 4. (a) Frequency shift of κ-casein layer upon exposure to 10 nM trypsin and plasmin. Exposure occurs at time = 0. The presence of only one active cleavage site on κ-casein results in similar mass changes by both trypsin and plasmin. (b) Frequency shift of κ-casein layer after 240 min of exposure to plasmin and trypsin at different crystal harmonics. Error bars correspond to amplitude of noise in (a).

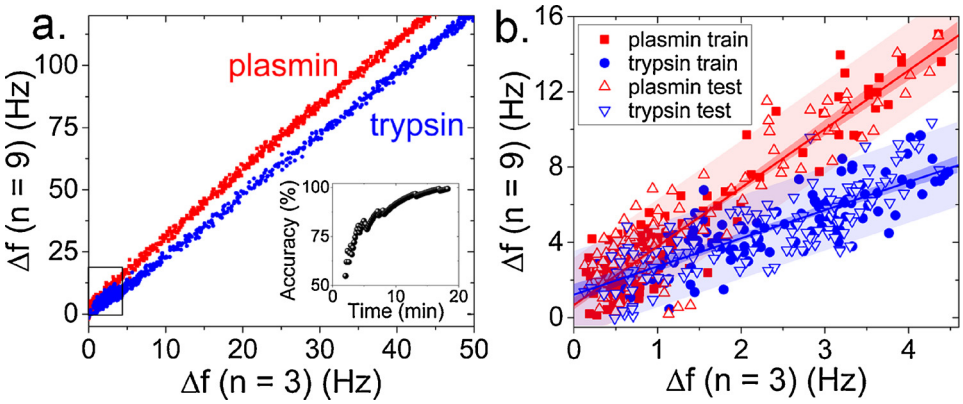

Fig. 5. Machine learning-based identification of protease using self-assembled κ-casein. a. Frequency shift of 9th vs. 3rd harmonics of κ-casein-coated QCM at during 10 nM plasmin and trypsin exposure. Inset shows accuracy of support vector classification of plasmin and trypsin as a function of plasmin/trypsin exposure time. b. Zoom of the boxed inset region in (a), showing linear regressions (solid lines), prediction bands (narrow dark shaded regions), and 95% confidence bands (wide light shaded regions). Legend shows points used for training (solid shapes) and testing the support vector regression used for distinguishing between trypsin and plasmin.

Although the effect of plasmin and trypsin on κ-casein was similar, we investigated the possibility of using machine learning to differentiate between activity of plasmin and trypsin based on multi-harmonic response of the κ-casein layer. Fig. 4a shows a clear difference between plasmin and trypsin responses in the 3rd harmonic frequency shifts from t = 50-150 min, and a significant difference between the plasmin and trypsin 9th harmonic responses at t > 150 min. To dmeonstrate differentiation between plasmin and trypsin, we plotted Δf at the 9th harmonic ($\Delta f_{n=9}$) vs Δf at the 3rd harmonic ($\Delta f_{n=3}$) during 10 nM plasmin and trypsin exposure (Fig. 5a). Response of the 5th, 7th, and 11th harmonics could be used as additional inputs for improving classification accuracy but were not included in order to simplify visualization of the 2-dimensional decision boundary. It is clear that after several hours of exposure to protease, corresponding to the upper half of Fig. 5a, the effect of trypsin on κ-casein can be easily distinguished from the effect of plasmin with 100% accuracy. To check classification accuracy for shorter detection times, we investigated the region shown in the small inset box in the lower left-hand corner of Fig. 5a, b shows the data and decision boundary located inside the inset box. Points used for training and testing a support vector regression (SVR) for distinguishing between trypsin and plasmin are described in the legend. Linear regressions and associated prediction bands and confidence bands demonstrate the possibility of accurate discrimination between trypsin and casein after frequency shift of the 3rd harmonic reaches ~3 Hz. Accuracy of the SVR as a function of trypsin/plasmin exposure time is shown in the inset of Fig. 5a. From t = 0–5 min, classification accuracy is low due to the highly localized set of points, likely because the casein film has not yet undergone significant proteolysis. From 15–20 min, the observations are classified with more than 95% accuracy. The result demonstrates that the κ-casein response to trypsin and plasmin may be used to perform accurate trypsin and plasmin identification after a detection time of ~20 min or less using statistical approaches.

4. Conclusions

Using multi-harmonic QCM frequency response, we demonstrated self-assembly of α-, β-, and κ-casein layers on hydrophilic SiO_2 without the use of crosslinking. While α-casein was removed upon flushing with PBS, β- and κ-casein layers remained immobilized under PBS flow. Using machine learning techniques, we showed that it is possible to achieve > 90% accurate identification of α-, β-, and κ-casein after 1 min of casein assembly on SiO_2. Frequency shift of the κ-casein layer was similar in response to proteolysis by both trypsin and plasmin due to the single cleavage site on κ-casein. The κ-casein layer enabled detection limits of 0.2 nM plasmin and 0.3 nM trypsin and allowed accurate discrimination between plasmin and trypsin after 15 min of 10 nM protease exposure. The β-casein layer showed higher mass loss when exposed to trypsin than plasmin, likely because β-casein contains at least one more cleavage site for trypsin than for plasmin. The β-casein layer allowed for 0.5 nM detection of plasmin and 0.2 nM detection of trypsin with accurate discrimination between plasmin and trypsin after 2 min of 1 nM protease exposure. Significant desorption of immobilized β-casein occurred during exposure to alkaline pH (pH > 8) environment due to accumulation of surface charge which leads to repulsion between casein molecules and between casein and SiO_2. Changes in cohesion/rigidity of β-casein layers that occurred in acidic environments (pH < 6) were reversible upon return to neutral pH. Results of the study demonstrate marker-free detection and identification of trypsin and plasmin proteases using self-assembled casein-based biosensors which achieve sensitivity comparable to conventional protease detection techniques.

Acknowledgement

This research was conducted at the Center for Nanophase Materials Sciences, which is a US Department of Energy Office of Science User Facility. MT, VS, TH acknowledge funding from the European Union's Horizon 2020 research and innovation program under the Marie Sklodowska-Curie grant agreement No. 690898. This manuscript has been authored by UT-Battelle, LLC under Contract No. DE-AC05-00OR22725 with the U.S. Department of Energy. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (http://energy.gov/downloads/doe-public-access-plan).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.snb.2018.05.100.

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